

Bangladesh University of Business & Technology (BUBT)



LAB Report

Course Code : CSE 210
Course Title : Operating Systems Lab
Experiment No. : 03

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Intake: 53
Section: 01
Program: B.Sc. Engg. in CSE

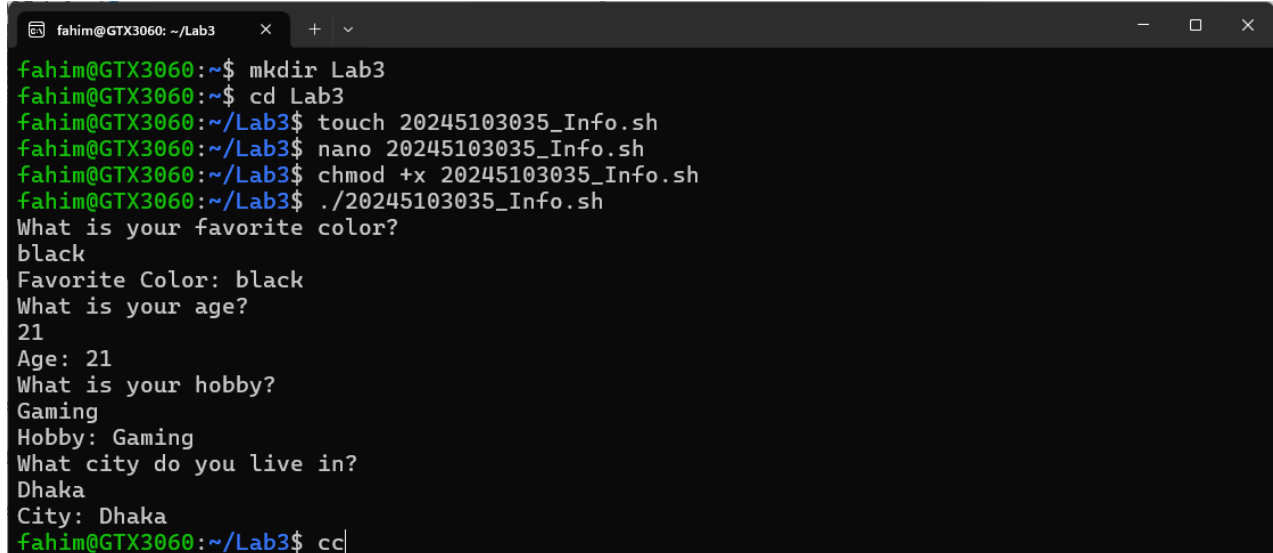
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Date of Submission: 18/08/2025

Signature of Teacher

1. Create a bash script named 20245103035_Info.sh

```
touch 20245103035_Info.sh
nano 20245103035_Info.sh
chmod +x 20245103035_Info.sh
./20245103035_Info.sh
```

A terminal window titled 'fahim@GTX3060: ~/Lab3' showing the execution of a bash script. The user runs 'mkdir Lab3', 'cd Lab3', 'touch 20245103035_Info.sh', 'nano 20245103035_Info.sh', 'chmod +x 20245103035_Info.sh', and finally './20245103035_Info.sh'. The script prompts for 'What is your favorite color?' (black), 'What is your age?' (21), 'What is your hobby?' (Gaming), and 'What city do you live in?' (Dhaka). The output shows 'Favorite Color: black', 'Age: 21', 'Hobby: Gaming', and 'City: Dhaka'. The prompt 'fahim@GTX3060:~/Lab3\$ cc|' is visible at the bottom.

```
fahim@GTX3060:~$ mkdir Lab3
fahim@GTX3060:~$ cd Lab3
fahim@GTX3060:~/Lab3$ touch 20245103035_Info.sh
fahim@GTX3060:~/Lab3$ nano 20245103035_Info.sh
fahim@GTX3060:~/Lab3$ chmod +x 20245103035_Info.sh
fahim@GTX3060:~/Lab3$ ./20245103035_Info.sh
What is your favorite color?
black
Favorite Color: black
What is your age?
21
Age: 21
What is your hobby?
Gaming
Hobby: Gaming
What city do you live in?
Dhaka
City: Dhaka
fahim@GTX3060:~/Lab3$ cc|
```

Inside 20245103035_Info.sh:

```
# ./bin/bash
```

```
echo "What is your favorite color?"
read a
echo "Favorite Color: $a"
```

```
echo "What is your age?"
read b
echo "Age: $b"
```

```
echo "What is your hobby?"
read c
echo "Hobby: $c"
```

```
echo "What city do you live in?"
read d
echo "City: $d"
```

```
fahim@GTX3060: ~/Lab3
GNU nano 7.2 20245103035_Info.sh *
# ./bin/bash

echo "What is your favorite color?"
read a
echo "Favorite Color: $a"

echo "What is your age?"
read b
echo "Age: $b"

echo "What is your hobby?"
read c
echo "Hobby: $c"

echo "What city do you live in?"
read d
echo "City: $d"
cc|

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
              ^_ Location
              ^/ Go To Line
```

2. Create a bash script named Operations.sh

```
touch 20245103035_Operations.sh
nano 20245103035_Operations.sh
chmod +x 20245103035_Operations.sh
./20245103035_Operations.sh
```

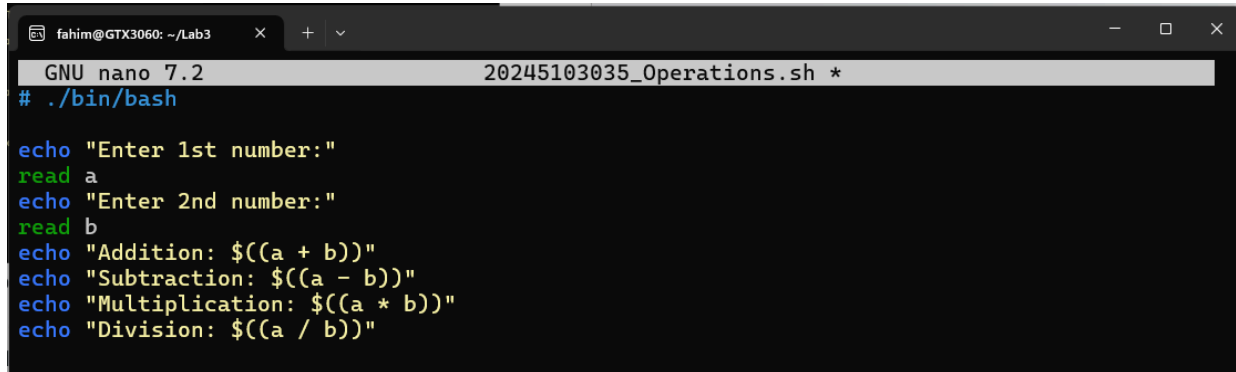
```
fahim@GTX3060: ~/Lab3
fahim@GTX3060:~/Lab3$ touch 20245103035_Operations.sh
fahim@GTX3060:~/Lab3$ nano 20245103035_Operations.sh
fahim@GTX3060:~/Lab3$ chmod +x 20245103035_Operations.sh
fahim@GTX3060:~/Lab3$ ./20245103035_Operations.sh
Enter 1st number:
10
Enter 2nd number:
5
Addition: 15
Subtraction: 5
Multiplication: 50
Division: 2
fahim@GTX3060:~/Lab3$ |
```

Inside 20245103035_Operations.sh:

```
# ./bin/bash
```

```
echo "Enter 1st number:"
read a
echo "Enter 2nd number:"
read b
echo "Addition: $((a + b))"
echo "Subtraction: $((a - b))"
```

```
echo "Multiplication: $((a * b))"
echo "Division: $((a / b))"
```

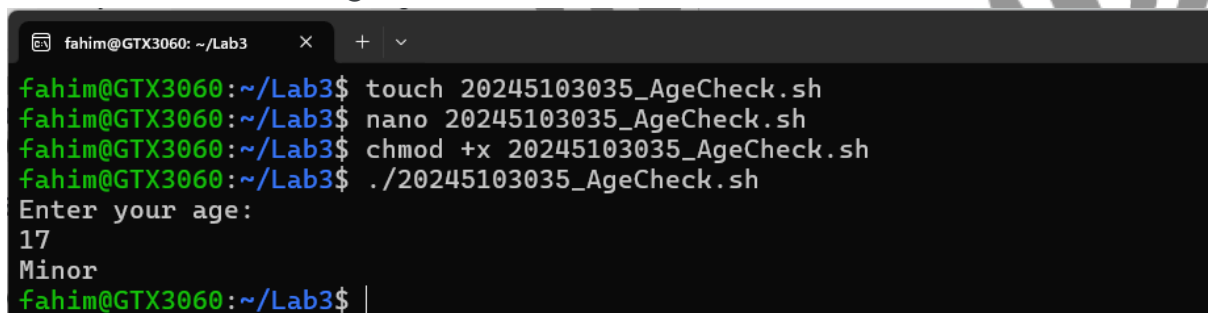


```
fahim@GTX3060: ~/Lab3
GNU nano 7.2 20245103035_Operations.sh *
# ./bin/bash

echo "Enter 1st number:"
read a
echo "Enter 2nd number:"
read b
echo "Addition: $((a + b))"
echo "Subtraction: $((a - b))"
echo "Multiplication: $((a * b))"
echo "Division: $((a / b))"
```

3. Create a bash script named 20245103035_AgeCheck.sh

```
touch 20245103035_AgeCheck.sh
nano 20245103035_AgeCheck.sh
chmod +x 20245103035_AgeCheck.sh
./20245103035_AgeCheck.sh
```



```
fahim@GTX3060: ~/Lab3
fahim@GTX3060:~/Lab3$ touch 20245103035_AgeCheck.sh
fahim@GTX3060:~/Lab3$ nano 20245103035_AgeCheck.sh
fahim@GTX3060:~/Lab3$ chmod +x 20245103035_AgeCheck.sh
fahim@GTX3060:~/Lab3$ ./20245103035_AgeCheck.sh
Enter your age:
17
Minor
fahim@GTX3060:~/Lab3$ |
```

Inside 20245103035_AgeCheck.sh:

```
# ./bin/bash
```

```
echo "Enter your age:"
read z
```

```
if [ $z -ge 18 ]; then
    echo "Adult"
else
    echo "Minor"
fi
```

```
fahim@GTX3060: ~/Lab3
GNU nano 7.2 20245103035_AgeCheck.sh *
# ./bin/bash

echo "Enter your age:"
read z

if [ $z -ge 18 ]; then
    echo "Adult"
else
    echo "Minor"
fi
```

4. Create a bash script named 20245103035_ArrayAccess.sh

```
touch 20245103035_ArrayAccess.sh
nano 20245103035_ArrayAccess.sh
chmod +x 20245103035_ArrayAccess.sh
./20245103035_ArrayAccess.sh
```

```
fahim@GTX3060: ~/Lab3
fahim@GTX3060:~/Lab3$ touch 20245103035_ArrayAccess.sh
fahim@GTX3060:~/Lab3$ nano 20245103035_ArrayAccess.sh
fahim@GTX3060:~/Lab3$ chmod +x 20245103035_ArrayAccess.sh
fahim@GTX3060:~/Lab3$ ./20245103035_ArrayAccess.sh
Enter index number (0, 1, 2 for name, age, city):1
Value at index : 21
fahim@GTX3060:~/Lab3$ |
```

Inside 20245103035_ArrayAccess.sh:

```
# ./bin/bash
```

```
Arr=(Fahim 21 Dhaka)
```

```
read -p "Enter index number (0, 1, 2 for name, age, city):" x
```

```
echo "Value at index $index: ${Arr[$x]}"
```

```
fahim@GTX3060: ~/Lab3
GNU nano 7.2 20245103035_ArrayAccess.sh *
# ./bin/bash

Arr=(Fahim 21 Dhaka)

read -p "Enter index number (0, 1, 2 for name, age, city):" x

echo "Value at index $index: ${Arr[$x]}"
```